

A Leavitt-type Banach algebra which is not a Calkin algebra

Expanded proof draft with quotient-norm details

Draft for verification

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Abstract

We construct a unital Banach algebra A which is not isomorphic, as a Banach algebra, to $\mathcal{B}(X)/\mathcal{K}(X)$ for any Banach space X . The algebra is a simple quotient of the norm-closed algebra generated by the prefix-shift representation on an ℓ_1 -space of the Leavitt path algebra of the one-vertex graph with $(2^{\aleph_0})^+$ loops. Separable Banach spaces are excluded by density. Nonseparable Banach spaces are excluded by an ℓ_1 matrix-unit obstruction: a nonseparable Banach space whose Calkin algebra is simple cannot contain countable matrix units with ℓ_1 row geometry. The proof below writes out the quotient-norm and separable-range details explicitly.

Conventions. All Banach spaces and Banach algebras are complex. A Banach algebra isomorphism means a bounded algebra isomorphism with bounded inverse. Ideals in Banach algebras are closed two-sided ideals unless explicitly called algebraic ideals. For a Banach space X , $\mathcal{B}(X)$ denotes the algebra of bounded operators on X , $\mathcal{K}(X)$ the ideal of compact operators, and $\mathcal{Q}(X) = \mathcal{B}(X)/\mathcal{K}(X)$ the Calkin algebra.

1 Two elementary quotient-norm facts

We begin with two small facts that will be used in the nonseparable obstruction. They are included to make explicit the functional-analytic bookkeeping that is often left implicit.

Lemma 1.1 (Induced operator norm from a corrected lift). *Let X be a Banach space, let $N \subseteq X$ be a closed subspace, and let $\pi : X \rightarrow X/N$ be the quotient map. Suppose $U, K \in \mathcal{B}(X)$ satisfy*

$$U(N) \subseteq N, \quad K(X) \subseteq N.$$

Then $U + K$ also leaves N invariant, and the operators induced by U and by $U + K$ on X/N are equal. Consequently

$$\|\bar{U}\| \leq \|U + K\|,$$

where $\bar{U}$ denotes the operator on X/N induced by U .

Proof. Since $K(X) \subseteq N$, in particular $K(N) \subseteq N$. Thus $U + K$ leaves N invariant. For every $x \in X$,

$$\pi((U + K)x) = \pi(Ux) + \pi(Kx) = \pi(Ux),$$

because $Kx \in N$. Hence the induced operators coincide. The norm of an induced operator is bounded by the norm of any bounded operator that induces it, so $\|\bar{U}\| \leq \|U + K\|$. $\square$

Lemma 1.2 (Absorbing countably many compact errors). *Let X be a Banach space, let $(R_n)_{n \in \mathbb{N}} \subseteq \mathcal{B}(X)$ be countable, and let $(K_m)_{m \in \mathbb{N}} \subseteq \mathcal{K}(X)$ be countable. There is a separable closed subspace $N \subseteq X$ such that*

$$K_m(X) \subseteq N \quad (m \in \mathbb{N}), \quad R_n(N) \subseteq N \quad (n \in \mathbb{N}).$$

Proof. For each compact K_m , the closed linear span of $K_m(X)$ is separable: indeed, $K_m(B_X)$ has compact norm closure, hence is a compact metric space and is separable, while

$$K_m(X) = \bigcup_{r=1}^{\infty} rK_m(B_X).$$

Let N_0 be the closed linear span of $\bigcup_m K_m(X)$. Then N_0 is separable. Let N be the closed linear span of all vectors of the form

$$R_{i_1} R_{i_2} \cdots R_{i_k} x,$$

where $k \geq 0$, $i_1, \dots, i_k \in \mathbb{N}$, and $x \in N_0$; for $k = 0$ this means $x \in N_0$. There are only countably many finite words in the countable alphabet $\{R_n : n \in \mathbb{N}\}$, and each image of the separable space N_0 is separable. Hence N is separable. Its definition also gives $R_n(N) \subseteq N$ for every n , and clearly $K_m(X) \subseteq N_0 \subseteq N$. $\square$

2 The Leavitt-type Banach algebra

Let

$$\mathfrak{c} = 2^{\aleph_0}, \quad \kappa = \mathfrak{c}^+.$$

Let E_κ be the directed graph with one vertex v and κ loops. Let $L_\kappa = L_{\mathbb{C}}(E_\kappa)$ be the Leavitt path algebra. Write the real edge corresponding to $\alpha < \kappa$ as s_α and the ghost edge as t_α .

For arbitrary graphs, the finite Cuntz-Krieger relation is imposed only at regular vertices, i.e. vertices which are neither sinks nor infinite emitters. Since the unique vertex of E_κ is an infinite emitter, the relation $v = \sum ee^*$ is not imposed at v ; see Goodearl [2, Section 1.4]. Thus the relevant relations include

$$t_\alpha s_\beta = \delta_{\alpha\beta} 1 \quad (\alpha, \beta < \kappa),$$

together with the usual source and range relations, which are trivial here because there is only one vertex.

We shall use the standard simplicity theorem for arbitrary-graph Leavitt path algebras: $L_K(E)$ is simple if and only if the only hereditary saturated subsets of E^0 are $\emptyset$ and E^0 , and E satisfies Condition (L), meaning that every cycle has an exit; see Abrams, Ara and Siles Molina [1, Theorem 2.9.1]. For E_κ , the only subsets of $E^0 = \{v\}$ are $\emptyset$ and $\{v\}$. Also every cycle has an exit: at the single vertex one can choose a loop different from the next edge of the cycle, since $\kappa \geq 2$. Therefore

$$L_\kappa \text{ is algebraically simple.}$$

Let W_κ be the set of finite words over the alphabet κ , including the empty word $\emptyset$, and put

$$Y_\kappa = \ell_1(W_\kappa).$$

For $\alpha < \kappa$ define $S_\alpha, T_\alpha \in \mathcal{B}(Y_\kappa)$ on the unit vector basis by

$$S_\alpha \delta_w = \delta_{\alpha w},$$

and

$$T_\alpha \delta_{\beta w} = \begin{cases} \delta_w, & \beta = \alpha, \\ 0, & \beta \neq \alpha, \end{cases} \quad T_\alpha \delta_\emptyset = 0.$$

Then

$$\|S_\alpha\| = \|T_\alpha\| = 1, \quad T_\alpha S_\beta = \delta_{\alpha\beta} I.$$

Hence there is a unital representation

$$\rho : L_\kappa \longrightarrow \mathcal{B}(Y_\kappa), \quad \rho(s_\alpha) = S_\alpha, \quad \rho(t_\alpha) = T_\alpha.$$

Since L_κ is algebraically simple and $\rho(1) = I \neq 0$, the kernel of ρ is a proper algebraic ideal, hence zero. Thus ρ is faithful.

Let

$$B_\kappa = \overline{\rho(L_\kappa)} \subseteq \mathcal{B}(Y_\kappa)$$

be the norm-closed unital Banach algebra generated by $\rho(L_\kappa)$.

Lemma 2.1. *There exists a maximal proper closed two-sided ideal M of B_κ .*

Proof. Partially order the proper closed two-sided ideals by inclusion. Let $(I_j)_{j \in J}$ be a chain of such ideals. The norm closure of $\bigcup_j I_j$ is a closed two-sided ideal. It is proper: if 1 belonged to this closure, then some element of $\bigcup_j I_j$ would be within distance < 1 of 1, hence would be invertible in B_κ ; the ideal containing that element would then be all of B_κ , contradiction. Zorn's lemma gives a maximal proper closed two-sided ideal. $\square$

Fix such an ideal M , and define

$$A_\kappa = B_\kappa / M.$$

Then A_κ is a unital Banach algebra with no nonzero proper closed two-sided ideals: the inverse image of any closed ideal of A_κ is a closed ideal of B_κ containing M .

3 Density of the quotient algebra

Lemma 3.1. *The algebraic core survives in the quotient:*

$$M \cap \rho(L_\kappa) = \{0\}.$$

Consequently the composite map $L_\kappa \xrightarrow{\rho} B_\kappa \rightarrow A_\kappa$ is injective.

Proof. Suppose $0 \neq a \in M \cap \rho(L_\kappa)$. Since $\rho(L_\kappa)$ is algebraically simple, the algebraic two-sided ideal generated by a inside $\rho(L_\kappa)$ is all of $\rho(L_\kappa)$. Thus there are $x_1, \dots, x_n, y_1, \dots, y_n \in \rho(L_\kappa)$ such that

$$1 = \sum_{r=1}^n x_r a y_r.$$

The right-hand side belongs to M , because M is a two-sided ideal of B_κ . Hence $1 \in M$, contradicting the properness of M . $\square$

For $\alpha < \kappa$, put

$$p_\alpha = s_\alpha t_\alpha \in L_\kappa,$$

and use the same notation for its image in A_κ . By Lemma 3.1, each p_α is nonzero in A_κ . The relations give pairwise orthogonal idempotents:

$$p_\alpha p_\beta = \delta_{\alpha\beta} p_\alpha.$$

If p, q are nonzero orthogonal idempotents in a normed algebra, then $\|p - q\| \geq 1$, because

$$p(p - q) = p$$

and hence

$$\|p\| \leq \|p\| \|p - q\|.$$

Since $\|p\| > 0$, division gives $\|p - q\| \geq 1$.

Thus A_κ contains a 1-separated set of cardinality κ , so

$$\text{dens}(A_\kappa) \geq \kappa.$$

For clarity, this last implication means that balls of radius $1/3$ around the p_α 's are pairwise disjoint, so any dense subset must meet each of them.

Conversely, L_κ has cardinality at most κ : its elements are finite complex linear combinations of finite words in κ many generators, and $\kappa \geq \mathfrak{c} = |\mathbb{C}|$. Since B_κ is the norm closure of $\rho(L_\kappa)$, we have $\text{dens}(B_\kappa) \leq \kappa$, and therefore $\text{dens}(A_\kappa) \leq \kappa$. Hence

$$\boxed{\text{dens}(A_\kappa) = \kappa = \mathfrak{c}^+}.$$

4 Separable Banach spaces are excluded

Lemma 4.1. *If X is separable, then $\text{dens}(\mathcal{Q}(X)) \leq \mathfrak{c}$.*

Proof. A separable Banach space has cardinality at most $\mathfrak{c}$. Fix a countable dense subset $D \subseteq X$. Every bounded operator $T \in \mathcal{B}(X)$ is determined by its restriction to D , since T is continuous. Therefore

$$|\mathcal{B}(X)| \leq |X|^{|D|} \leq \mathfrak{c}^{\aleph_0} = \mathfrak{c}.$$

It follows that

$$\text{dens}(\mathcal{Q}(X)) \leq |\mathcal{Q}(X)| \leq |\mathcal{B}(X)| \leq \mathfrak{c}.$$

□

Since $\text{dens}(A_\kappa) = \mathfrak{c}^+ > \mathfrak{c}$, the algebra A_κ is not isomorphic, as a Banach algebra, to $\mathcal{Q}(X)$ for any separable Banach space X .

5 The nonseparable obstruction

The next lemma is the heart of the proof. It has been written in detail to make every quotient-norm and separability step explicit.

Lemma 5.1 (The ℓ_1 matrix-unit obstruction). *Let X be a nonseparable Banach space. Suppose $\mathcal{Q}(X)$ has no nonzero proper closed two-sided ideals. Then $\mathcal{Q}(X)$ cannot contain elements $(e_{ij})_{i,j \in \mathbb{N}}$ satisfying:*

1. $e_{ij}e_{kl} = \delta_{jk}e_{il}$ for all i, j, k, l ;
2. $e_{11} \neq 0$;
3. there is $C_r < \infty$ such that, for every finite $F \subseteq \mathbb{N}$ and every scalar family $(a_j)_{j \in F}$,

$$\left\| \sum_{j \in F} a_j e_{1j} \right\| \leq C_r \sup_{j \in F} |a_j|;$$

4. there is $C_c < \infty$ such that $\sup_i \|e_{i1}\| \leq C_c$.

Proof. Let $q : \mathcal{B}(X) \rightarrow \mathcal{Q}(X)$ be the quotient map.

Step 1: the separable-range ideal collapses to the compact ideal. Define

$$\mathcal{S}(X) = \{T \in \mathcal{B}(X) : \overline{T(X)} \text{ is separable}\}.$$

This is a two-sided ideal of $\mathcal{B}(X)$: if T has separable range closure, then ATB has range contained in $A(\overline{T(X)})$ for $A, B \in \mathcal{B}(X)$, hence also has separable range closure.

It is closed in norm. Indeed, if $T_n \rightarrow T$ and every $\overline{T_n(X)}$ is separable, then for every $x \in X$,

$$Tx = \lim_{n \rightarrow \infty} T_n x,$$

so $T(X)$ is contained in the closed linear span of $\bigcup_n T_n(X)$, a separable subspace.

Also $\mathcal{K}(X) \subseteq \mathcal{S}(X)$. If $C \in \mathcal{K}(X)$, then $\overline{C(B_X)}$ is compact and metric, hence separable, and

$$C(X) = \bigcup_{m=1}^{\infty} mC(B_X),$$

so $\overline{C(X)}$ is separable. Finally, $\mathcal{S}(X) \neq \mathcal{B}(X)$ because X is nonseparable and $I_X(X) = X$.

Since $\mathcal{Q}(X)$ is simple as a Banach algebra, the closed ideal $\mathcal{S}(X)/\mathcal{K}(X)$ of $\mathcal{Q}(X)$ is either 0 or all of $\mathcal{Q}(X)$. It is not all of $\mathcal{Q}(X)$, because its inverse image under q is $\mathcal{S}(X) \neq \mathcal{B}(X)$. Thus

$$\boxed{\mathcal{S}(X) = \mathcal{K}(X)}. \tag{5.1}$$

Consequently every separable-range operator on X is compact.

Step 2: lift the matrix units and absorb all compact defects into a separable invariant subspace. Choose arbitrary lifts $E_{ij} \in \mathcal{B}(X)$ with $q(E_{ij}) = e_{ij}$. The matrix-unit relations in $\mathcal{Q}(X)$ say that

$$D_{ij,kl} := E_{ij}E_{kl} - \delta_{jk}E_{il} \in \mathcal{K}(X)$$

for all i, j, k, l .

For each finite $F \subseteq \mathbb{N}$ and each rational complex tuple $a = (a_j)_{j \in F} \in (\mathbb{Q} + i\mathbb{Q})^F$, set

$$U_{F,a} = \sum_{j \in F} a_j E_{1j}.$$

Since

$$\|q(U_{F,a})\| = \left\| \sum_{j \in F} a_j e_{1j} \right\| \leq C_r \sup_{j \in F} |a_j|,$$

the definition of the quotient norm gives a compact operator $C_{F,a} \in \mathcal{K}(X)$ such that

$$\|U_{F,a} + C_{F,a}\| \leq (C_r + 1) \sup_{j \in F} |a_j|. \quad (5.2)$$

Likewise, for each i choose $H_i \in \mathcal{K}(X)$ such that

$$\|E_{i1} + H_i\| \leq C_c + 1. \quad (5.3)$$

The family consisting of all $D_{ij,kl}$, all $C_{F,a}$, and all H_i is countable. By Lemma 1.2, applied to the countable family (E_{ij}) and to this countable family of compact operators, there is a separable closed subspace $N \subseteq X$ such that

$$E_{ij}(N) \subseteq N \quad (i, j \in \mathbb{N}),$$

and the ranges of all $D_{ij,kl}$, $C_{F,a}$ and H_i are contained in N .

Let

$$Z = X/N,$$

let $\pi : X \rightarrow Z$ be the quotient map, and let $\bar{E}_{ij} \in \mathcal{B}(Z)$ be the operator induced by E_{ij} . Since $D_{ij,kl}(X) \subseteq N$, the induced operators satisfy exact matrix-unit relations:

$$\bar{E}_{ij} \bar{E}_{kl} = \delta_{jk} \bar{E}_{il}. \quad (5.4)$$

By Lemma 1.1, equation (5.2) implies, first for rational tuples and then by density for arbitrary finite scalar tuples, that

$$\left\| \sum_{j \in F} a_j \bar{E}_{1j} \right\| \leq (C_r + 1) \sup_{j \in F} |a_j|. \quad (5.5)$$

Similarly, (5.3) implies

$$\sup_i \|\bar{E}_{i1}\| \leq C_c + 1. \quad (5.6)$$

Here the point is that the compact correctors have range in N , so they induce the zero operator on Z .

Step 3: the first corner remains nonzero on Z . We claim that

$$\bar{E}_{11} \neq 0. \quad (5.7)$$

If $\bar{E}_{11} = 0$, then $E_{11}(X) \subseteq N$, so E_{11} has separable range. By (5.1), $E_{11} \in \mathcal{K}(X)$, whence $e_{11} = q(E_{11}) = 0$, contradicting the hypothesis.

Choose $y \in Z$ such that $\bar{E}_{11}y \neq 0$, and put

$$z_0 = \bar{E}_{11}y.$$

Then $z_0 \neq 0$ and, by idempotence of $\bar{E}_{11}$,

$$\bar{E}_{11}z_0 = z_0. \quad (5.8)$$

By the Hahn-Banach theorem, choose $\varphi \in Z^*$ with

$$\varphi(z_0) = 1. \quad (5.9)$$

Step 4: construct a bounded noncompact map $X \rightarrow \ell_1$. Define

$$\Theta : X \longrightarrow \ell_1$$

by

$$\Theta x = (\varphi(\bar{E}_{1j}\pi x))_{j=1}^\infty. \quad (5.10)$$

To see that this is well-defined and bounded, fix $x \in X$ and $n \in \mathbb{N}$. In the complex scalar case,

$$\sum_{j=1}^n |b_j| = \sup_{|a_j| \leq 1} \left| \sum_{j=1}^n a_j b_j \right|.$$

Using this and (5.5),

$$\begin{aligned} \sum_{j=1}^n |\varphi(\bar{E}_{1j}\pi x)| &= \sup_{|a_j| \leq 1} \left| \varphi \left(\sum_{j=1}^n a_j \bar{E}_{1j}\pi x \right) \right| \\ &\leq (C_r + 1) \|\varphi\| \|\pi x\| \\ &\leq (C_r + 1) \|\varphi\| \|x\|. \end{aligned}$$

Taking the supremum over n shows $\Theta x \in \ell_1$ and

$$\|\Theta\| \leq (C_r + 1) \|\varphi\|.$$

For $i \in \mathbb{N}$, put

$$z_i = \bar{E}_{i1} z_0.$$

By (5.6), the sequence (z_i) is bounded:

$$\|z_i\| \leq (C_c + 1) \|z_0\|.$$

Choose lifts $x_i \in X$ with $\pi x_i = z_i$ and

$$\|x_i\| \leq \|z_i\| + 1.$$

Then (x_i) is bounded. For all $i, k \in \mathbb{N}$, equations (5.4), (5.8), and (5.9) give

$$\varphi(\bar{E}_{1k} z_i) = \varphi(\bar{E}_{1k} \bar{E}_{i1} z_0) = \delta_{ki} \varphi(\bar{E}_{11} z_0) = \delta_{ki}.$$

Therefore

$$\Theta x_i = u_i,$$

where (u_i) is the unit vector basis of ℓ_1 . Since (u_i) has no norm-convergent subsequence, Θ is not compact.

Step 5: turn this into a noncompact separable-range operator on X . Since X is nonseparable, it is infinite-dimensional. By Riesz's lemma, choose a bounded sequence $(w_i) \subseteq X$ with no

norm-convergent subsequence; for instance, one may choose unit vectors separated by a fixed positive distance. Define

$$R : \ell_1 \longrightarrow X, \quad R((a_i)) = \sum_{i=1}^{\infty} a_i w_i.$$

This is bounded, with $\|R\| \leq \sup_i \|w_i\|$. Let

$$T = R\Theta : X \longrightarrow X.$$

Then

$$T(X) \subseteq \overline{\text{span}}\{w_i : i \in \mathbb{N}\},$$

so T has separable range. However,

$$Tx_i = R\Theta x_i = Ru_i = w_i,$$

and (w_i) has no norm-convergent subsequence while (x_i) is bounded. Hence T is not compact. This contradicts (5.1). The lemma follows. $\square$

6 A_κ contains the forbidden matrix units

Choose countably many distinct loops

$$\alpha_1, \alpha_2, \dots < \kappa.$$

In A_κ , define

$$e_{ij} = \text{the image of } s_{\alpha_i} t_{\alpha_j}.$$

By Lemma 3.1, $e_{11} \neq 0$. The Leavitt relations give

$$e_{ij} e_{kl} = \delta_{jk} e_{il}.$$

We verify the ℓ_1 row estimate. In the concrete representation on $Y_\kappa = \ell_1(W_\kappa)$, set

$$U = \sum_{j \in F} a_j S_{\alpha_1} T_{\alpha_j}$$

for finite $F \subseteq \mathbb{N}$. If $x = (c_w)_{w \in W_\kappa} \in \ell_1(W_\kappa)$, then the coefficient of Ux at the basis vector $\delta_{\alpha_1 w}$ is

$$\sum_{j \in F} a_j c_{\alpha_j w},$$

and all other coefficients are zero. Hence

$$\begin{aligned} \|Ux\|_1 &= \sum_{w \in W_\kappa} \left| \sum_{j \in F} a_j c_{\alpha_j w} \right| \\ &\leq \sup_{j \in F} |a_j| \sum_{w \in W_\kappa} \sum_{j \in F} |c_{\alpha_j w}| \\ &\leq \sup_{j \in F} |a_j| \|x\|_1. \end{aligned}$$

Therefore

$$\left\| \sum_{j \in F} a_j S_{\alpha_1} T_{\alpha_j} \right\| \leq \sup_{j \in F} |a_j|.$$

The quotient map $B_\kappa \rightarrow A_\kappa$ is contractive, so

$$\left\| \sum_{j \in F} a_j e_{1j} \right\| \leq \sup_{j \in F} |a_j|. \quad (6.1)$$

Likewise,

$$\sup_i \|e_{i1}\| \leq 1, \quad (6.2)$$

because $\|S_{\alpha_i} T_{\alpha_1}\| \leq 1$ for every i .

Thus A_κ contains a matrix-unit system satisfying all hypotheses of Lemma 5.1.

7 Conclusion

Theorem 7.1. *The unital Banach algebra*

$$A_\kappa = B_\kappa / M, \quad \kappa = (2^{\aleph_0})^+,$$

constructed above is not isomorphic, as a Banach algebra, to $\mathcal{B}(X)/\mathcal{K}(X)$ for any Banach space X .

Proof. Assume, toward contradiction, that

$$A_\kappa \cong \mathcal{Q}(X)$$

for some Banach space X , where the isomorphism is bounded with bounded inverse.

If X is separable, Lemma 4.1 gives

$$\text{dens}(\mathcal{Q}(X)) \leq \mathfrak{c} < \mathfrak{c}^+ = \text{dens}(A_\kappa),$$

which is impossible for topologically isomorphic Banach algebras.

Suppose X is nonseparable. Since A_κ has no nonzero proper closed two-sided ideals, neither does $\mathcal{Q}(X)$. Let $\Phi : A_\kappa \rightarrow \mathcal{Q}(X)$ be the isomorphism. The elements $\Phi(e_{ij})$ are matrix units in $\mathcal{Q}(X)$, and $\Phi(e_{11}) \neq 0$. By (6.1),

$$\left\| \sum_{j \in F} a_j \Phi(e_{1j}) \right\| \leq \|\Phi\| \sup_{j \in F} |a_j|,$$

and by (6.2),

$$\sup_i \|\Phi(e_{i1})\| \leq \|\Phi\|.$$

Thus the hypotheses of Lemma 5.1 hold in $\mathcal{Q}(X)$, a contradiction.

Therefore no Banach space X satisfies $A_\kappa \cong \mathcal{B}(X)/\mathcal{K}(X)$. □

Audit checklist

The proof rests on the following points, all made explicit above.

1. The Leavitt path algebra L_κ is algebraically simple by the arbitrary-graph simplicity theorem: the graph has one vertex, hence no nontrivial hereditary saturated subsets, and every cycle has an exit.
2. The maximal closed ideal quotient does not kill the algebraic Leavitt algebra: $M \cap \rho(L_\kappa) = 0$ by algebraic simplicity.
3. The density of A_κ is exactly κ : the upper bound comes from the dense algebraic subalgebra of cardinality κ , and the lower bound from κ many pairwise 1-separated nonzero orthogonal idempotents.
4. For nonseparable X , the separable-range operators form a closed proper ideal $\mathcal{S}(X)$ containing $\mathcal{K}(X)$. If $\mathcal{Q}(X)$ is simple, this forces $\mathcal{S}(X) = \mathcal{K}(X)$.
5. In Lemma 5.1, only countably many compact defect operators and compact norm-correctors are used. Their ranges are absorbed into a separable invariant subspace N .
6. The quotient-norm estimates on X/N follow from Lemma 1.1: compact correctors with range in N induce the zero operator on X/N .
7. The ℓ_1 row estimate is exactly what makes $\Theta : X \rightarrow \ell_1$ bounded. The column bound provides bounded vectors x_i with $\Theta x_i = u_i$. Composing with a bounded noncompact map $\ell_1 \rightarrow X$ gives a noncompact separable-range operator, contradicting $\mathcal{S}(X) = \mathcal{K}(X)$.

References

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